

TOPIC 3.3

How does energy transformation affect global systems?

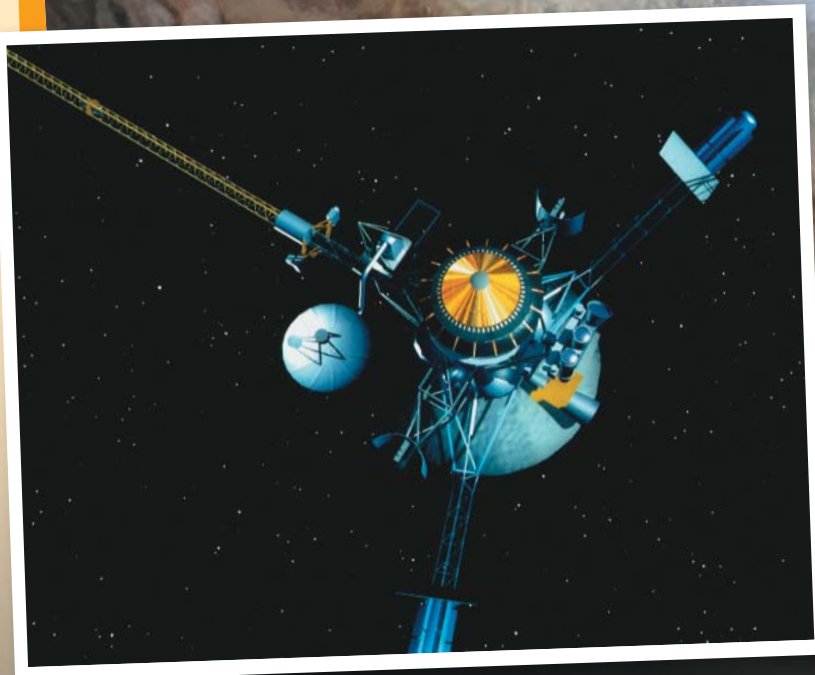
Key Concepts

- Earth is a system in which energy is transformed.
- Earth's atmosphere is heated by the transformation and transfer of solar energy and thermal energy.
- Energy transfer and energy transformation moderate Earth's temperature.
- Energy transformation and transfer can harm aquatic and terrestrial ecosystems.

Curricular Competencies

- Construct, analyze, and interpret graphs, models, and diagrams.
- Transfer and apply learning to new situations.
- Contribute to finding solutions to problems at the local and/or global level.

Knowing how energy is transformed helps us understand the processes that shape the unique environments of different planets. In 2003, NASA sent the space craft *Galileo* into the swirling, gaseous depths of Jupiter's atmosphere. The data collected by the spacecraft supported the theory that the gas giant's own gravitational potential energy is causing it to shrink. In the process, gravitational potential energy is being transformed into thermal energy. This thermal energy is the driving force behind intense storms in Jupiter's atmosphere, including a giant storm known as the "Great Red Spot." The diameter of several Earths put together, this super storm has been raging for nearly 200 years, perhaps longer.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Applying** How might gravitational potential energy be transformed into thermal energy as Jupiter shrinks?
2. **Inferring** Storms in Earth's atmosphere are also driven by energy transformations. What forms of energy might be involved in these transformations? (Hint: Where on Earth do the most severe storms occur? Is there a connection between these locations and the energy of the storms?)
3. **Making Connections** There is nowhere to recharge batteries in outer space. What energy transformation might have powered the *Galileo* spacecraft on its mission?
4. **Applying First Peoples Perspectives** First Peoples science is based on knowledge of the local environment. In what ways do energy transformations on a global scale impact life on a local scale? How do global energy transformations influence our identity and sense of place?



Key Terms

There are three key terms that are highlighted in bold type in this Topic:

- conduction
- convection
- specific heat capacity

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

Earth is a system in which energy is transformed.

Activity

Energy and Earth's Systems



- 1 “How do Earth’s systems rely on energy transformation?” In this activity, you will work with a partner to answer this question.
2. With your partner, brainstorm how living and nonliving processes on Earth involve energy transformations. Have one student act as recorder. The following questions can help you complete this task.
What energy transformations occur when
 - a plant carries out photosynthesis?
 - you eat a sandwich?
 - you sweat?
 - it rains?
 - a volcano erupts?
 - the wind blows?
 - the tide goes in and out?
3. Discuss the question from step 1 with your partner. Use your answers from step 2 to guide your discussion.
4. When you are done, share your ideas with another pair of partners, or with the class.

Consider the different processes that occur on Earth. These include Earth’s wind and weather, tides and currents, volcanic and mountain building processes, and the processes that take place in living things. None of these would occur without the transformation of energy. Topic 3.1 introduced the idea of a system. Recall that a system is anything that is being observed. This means that Earth, shown in **Figure 3.33**, can be considered a system, and often is. Three main types of energy play a role in this system: solar energy, gravitational potential energy, and nuclear energy.

Figure 3.33 Earth can be considered a system. All of Earth’s processes rely on energy transformation.

Inferring: What energy transformation drives the hurricane shown in the Atlantic Ocean in this figure?





Solar energy: The transformation of solar energy drives movement of water in the atmosphere and hydrosphere, as well as through and over land. It also provides the energy that sustains most life on Earth. All three of these effects are represented here in the South Okanagan Valley at Naramata.

Nuclear energy: Nuclear energy is transformed into thermal energy by radioactive decay in Earth's crust and mantle. This thermal energy causes volcanic activity, tectonic plate movement, geysers, and hot springs such as those at the village of Radium Hot Springs.



Gravitational potential energy: The transformation of this form of energy also helps to move water. Transformation of the Moon's gravitational potential energy causes the tides. Shown here is low tide at Long Beach on Vancouver Island.

Of these three types of energy—solar, gravitational potential, and nuclear—it is transformation of solar energy that has the greatest impact. It adds 5000 times more energy to the Earth system than nuclear decay. In total, about 174 000 terrajoules of energy move through the Earth system at any time. A Boeing 737 airplane would have to cross the Atlantic Ocean 174 000 trillion times to use this much energy. It is also 10 000 times as much energy as we generate on Earth in a given moment.



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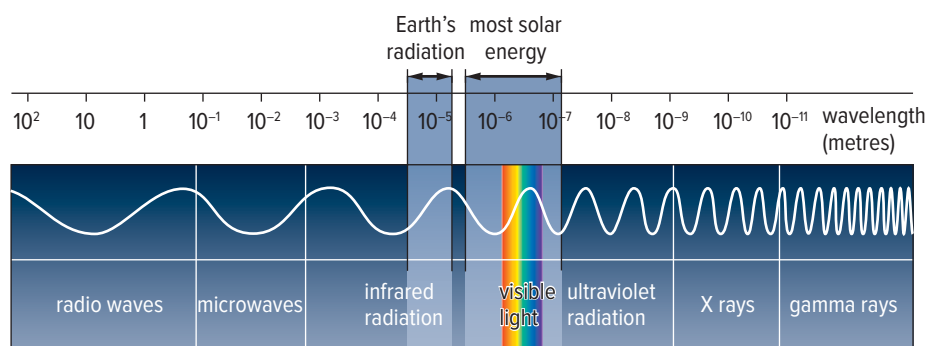
1. What types of energy contribute to the Earth system?
2. Which type of energy transformation has the greatest impact on Earth?

CONCEPT 2

Earth's atmosphere is heated by the transformation and transfer of solar energy and thermal energy.

Solar energy is made up of different wavelengths of electromagnetic radiation (**Figure 3.34**). These waves are transformed into thermal energy when they interact with matter, such as the ground, water, or air. The Sun is constantly radiating solar energy. Yet, Earth's temperature has remained relatively constant. This has provided conditions that are well-suited for life for millions of years. How is this possible? Following the path that energy takes through Earth's atmosphere in **Figure 3.35** provides an answer.

Figure 3.34 Electromagnetic radiation consists of the seven types shown here, from long-wavelength radio waves to short-wavelength gamma rays.



Absorption, Reflection, and Radiation of Energy

Most of the solar energy that reaches Earth's surface consists of visible light, infrared radiation, and ultraviolet radiation. Solar energy takes a variety of paths as it travels through Earth's atmosphere, as shown in **Figure 3.35**.

- About half of the solar energy that enters Earth's atmosphere is absorbed by the land and ocean. Visible light tends to pass through the atmosphere and reach Earth's surface unchanged. When it is absorbed by the surface, it is transformed into thermal energy. This energy is transformed and radiated into its surroundings as infrared radiation. The atmosphere absorbs much of the infrared radiation coming from Earth's surface. It temporarily traps some of this energy as thermal energy in the rapid motion of its molecules. This plays an important role in atmospheric warming, because these molecules constantly re-radiate the energy in all directions.
- Most of the remaining solar energy is absorbed, reflected, and scattered by clouds and particles in the atmosphere. Reflection and scattering involve energy transfer.
- This leaves about 10 percent that is reflected by Earth's surface.

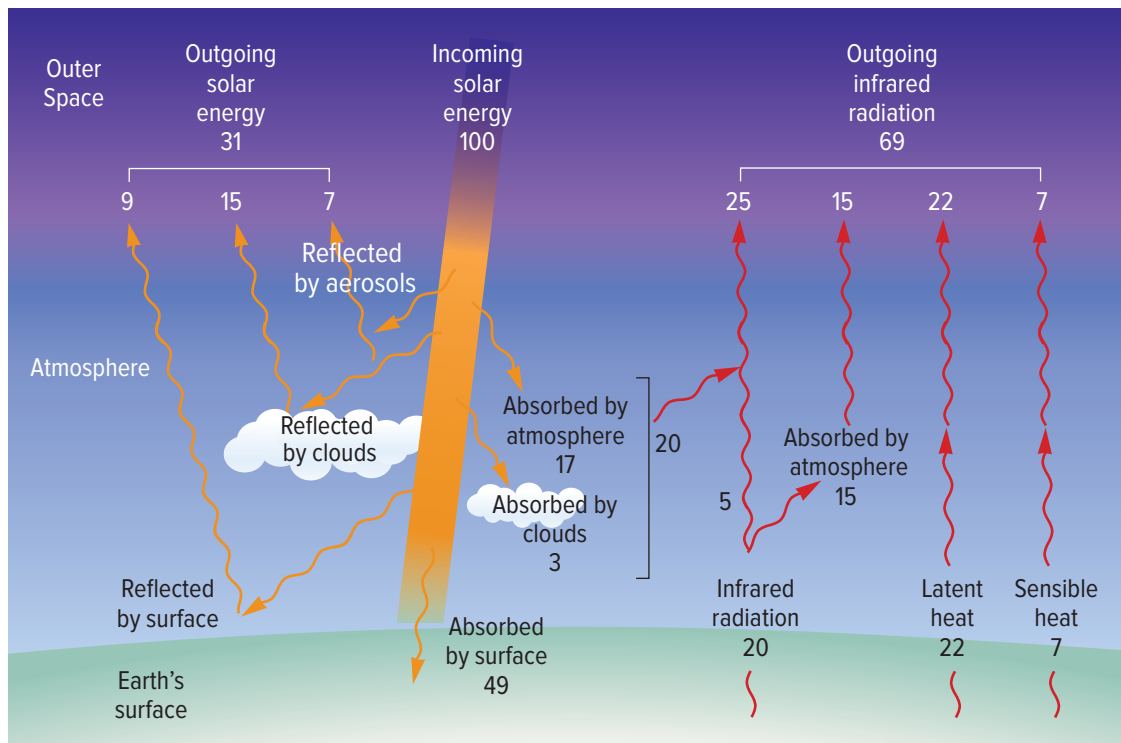


Figure 3.35 Incoming solar energy interacts with Earth's atmosphere and surface in different ways. This energy diagram is based on 100 units of solar energy entering the atmosphere. It shows the approximate amounts of energy radiated, reflected, and absorbed.

Activity

Analyzing Incoming and Outgoing Energy

Use **Figure 3.35** to answer the questions below with your partner or group.

1. What type of energy do the orange arrows represent?
2. What type of energy do the red arrows represent?
3. How can you verify that incoming and outgoing energy balance out?
4. Look up the terms *sensible* and *latent*. Record their meanings. Based on their meanings, what do you think the terms *sensible heat* and *latent heat* refer to in the lower right-hand corner of the figure?
5. a) How much solar energy is absorbed by the surface?
b) How much energy is radiated from the surface?
6. What percentage of incoming solar energy leaves Earth again over time?
7. Write a paragraph that describes a possible journey of the energy of a photon of solar energy through Earth's atmosphere.



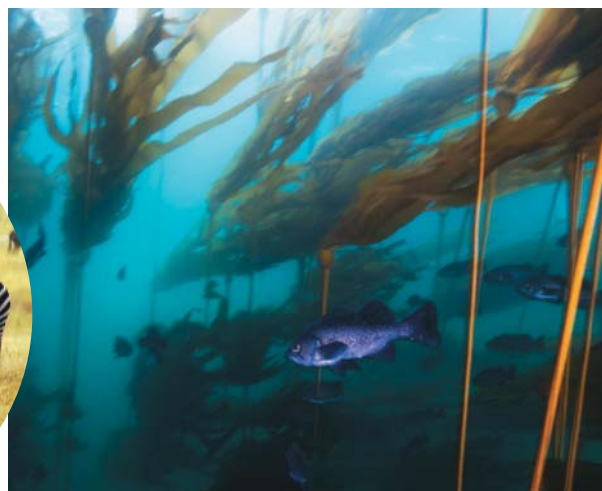
The Role of Greenhouse Gases

Greenhouse gases play an important role in warming Earth's atmosphere. Ones that are naturally present, such as carbon dioxide, water vapour, nitrous oxide (dinitrogen monoxide), and methane, are especially good at absorbing infrared radiation. The warming influence of these gases is often called the *greenhouse effect*. The greenhouse effect is the main reason

Earth's average temperature is about 14°C , instead of a very chilly -18°C . Earth's moderate average temperature allows life as we know it to thrive (**Figure 3.36**). In a later Concept, you will learn how human emissions of greenhouse gases has affected global ecosystems.



Figure 3.36 Ecosystems such as these could not exist if greenhouse gases were not present in Earth's atmosphere. **Applying:** Consider an ecosystem in your region. How would it change if there were no greenhouse gases in the atmosphere?



Conduction and Convection

You just read how solar energy is absorbed by Earth's surface and transformed into thermal energy. This thermal energy can be radiated again as infrared radiation. It can also be transferred to the atmosphere by conduction and convection. These processes are shown in **Figure 3.37**.

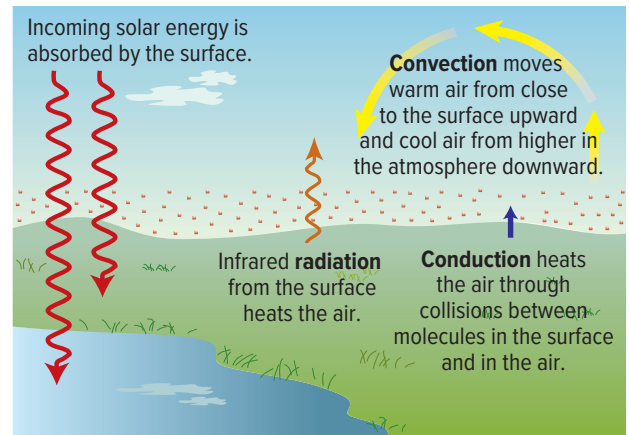
conduction the transfer of thermal energy between two substances that are touching

convection the transfer of thermal energy by the movement of heated fluids from one place to another

- **Conduction** is the transfer of thermal energy between two substances that are touching. After land and water absorb solar energy, their molecules collide more frequently with air molecules close to the surface. These collisions transfer thermal energy from the warmer surface to the cooler air by conduction. As a result, the temperature of the lower air comes close to the temperature of the land and water beneath it.
- **Convection** is the transfer of thermal energy by the movement of heated fluids (gases and liquids) from one place to another. Convection occurs as air circulates and distributes thermal energy. As the lower layer of the air warms, it expands, becoming less dense. Less-dense fluids rise, and more-dense fluids fall. As the cooler air falls, it takes the place of the rising warmer air.

Despite the important roles of conduction and convection, infrared radiation still plays the largest role in maintaining Earth's average temperature. Not only does radiation help warm the atmosphere, it also keeps it from getting too hot. Regardless of what processes solar energy undergoes once it reaches Earth, it is all eventually radiated back into space. This is shown back on **Figure 3.35**.

Figure 3.37 Radiation, conduction, and convection all affect atmospheric temperature.



Activity

Modelling Convection

In this activity, you will make a lava lamp to model convection in the atmosphere.

1. Add water to a 1 L clear plastic bottle with a funnel until it is three-quarters full.
2. Using the funnel, add vegetable oil until the bottle is nearly full.
3. Wait for the oil and water to separate. Then add 10 drops of food colouring to the bottle.
4. Add half of an effervescent tablet to the bottle. Observe what happens.
5. Your teacher may dim the lights. Add another half a tablet to the bottle. Then shine a flashlight through the bottom of the bottle to better observe what is happening.
6. **a)** How does this activity model convection in the atmosphere well?
b) What are the shortcomings of this model?
7. A real lava lamp has a heat source at the bottom of the lamp. This warms a waxy substance that expands and rises, carrying heat to the top of the lamp. When the wax rises, it cools again and sinks to the bottom of the lamp.
 - a)** How is this process like convection in the atmosphere?
 - b)** How is it different?



Before you leave this page . . .

1. Describe the roles played by the following in warming Earth's atmosphere.

a) radiation	c) greenhouse gases
b) conduction	d) convection

CONCEPT 3

Energy transfer and energy transformation moderate Earth's temperature.

Figure 3.38 The water cycle transports water and energy.

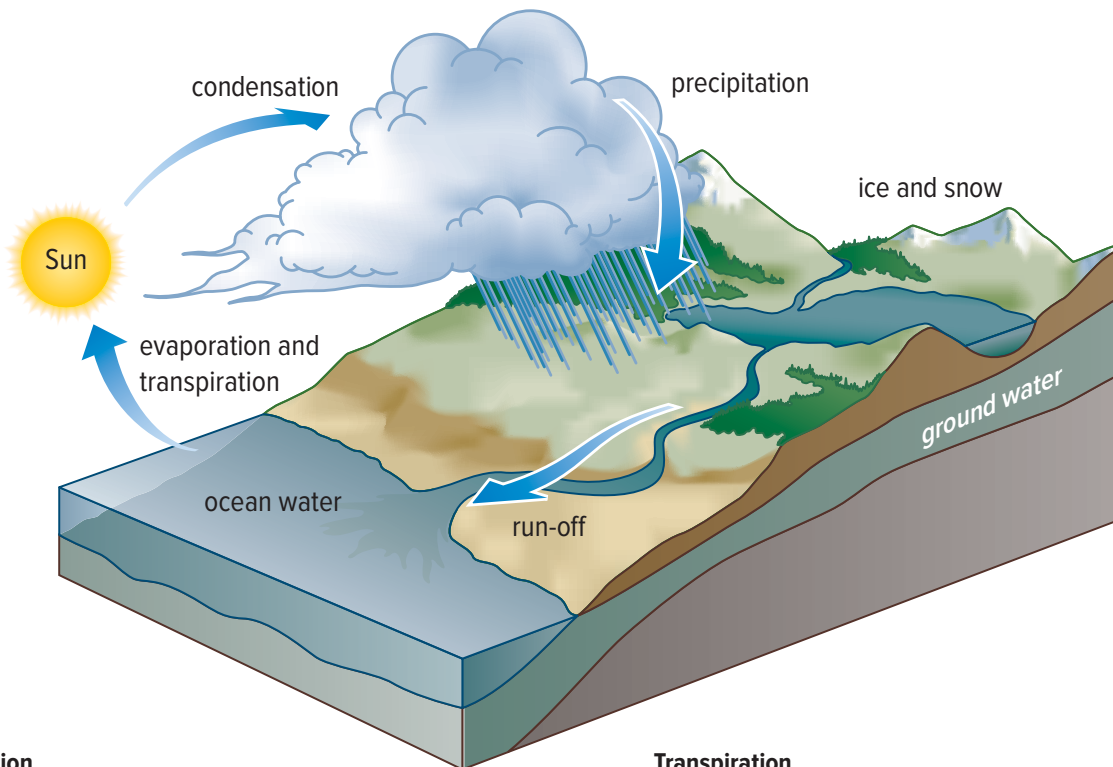
When scientists search for life on other planets, they often look for the presence of water. Water is the medium in which life processes, at least as we know them, occur. On Earth, water moves between three spheres—the hydrosphere, geosphere, and atmosphere—in the water cycle. It does so through the processes described below. As each of these processes transports water, they also move energy between Earth's different spheres, as shown in **Figure 3.38**.

Condensation

As water vapour rises in the atmosphere, it gains gravitational potential energy. It also releases thermal energy to air molecules through collisions. It then cools, condenses around particles in the atmosphere, and forms clouds. The condensation process releases more thermal energy to the atmosphere.

Precipitation

The water returns to Earth's surface through precipitation when it rains or snows. As the precipitation falls, its gravitational potential energy is transformed into mechanical kinetic energy. This transformation continues as water flows from higher to lower elevations on land. Thermal energy is also transferred to the atmosphere through friction as precipitation falls.



Evaporation

When water absorbs the Sun's energy, its molecules begin to move faster as the solar energy is transformed into thermal energy. If enough solar energy is transformed, water evaporates from Earth's surface. It enters the atmosphere as water vapour, bringing its thermal energy with it.

Transpiration

In transpiration, plants take up water from the ground, or bodies of water in the case of aquatic plants, and release it into the atmosphere as vapour. Like evaporation, transpiration moves thermal energy into the atmosphere.

Water Moderates Earth's Temperature

Water absorbs a lot of energy when it evaporates. As a result, evaporation has a cooling effect. You experience this when you sweat on a hot day. Your body temperature decreases as water evaporates from your skin. You notice this effect more if you stand in front of a fan, because the moving air increases the rate of evaporation from your skin. Evaporation cools Earth in a similar way, as it moves thermal energy away from the surface.

Storing energy also affects Earth's temperature. The amount of energy needed to change the temperature of 1 g of a substance by 1°C is called **specific heat capacity**. Water has a very high specific heat capacity, so Earth's vast oceans can store a large amount of thermal energy with very little change in temperature. Oceans also transfer and radiate energy to their cooler surroundings, but this happens slowly. Without this effect, Earth would experience much greater shifts in temperature between day and night. It would also be much colder in winter and warmer in summer.

The moderating effect of oceans is best seen in coastal regions. Water has a much higher specific heat capacity than land does. Thus, in summer, when land and water absorb the same amount of heat, the temperature of the land increases much more than that of water. In winter, when land and water lose the same amount of heat, land cools much more than water. For coastal regions, water acts like an air conditioner or heater during different seasons. Summers tend to be cooler compared with inland locations, and winters tend to be warmer.

specific heat capacity the amount of energy required to change the temperature of 1 g of a substance by 1°C

Connect to Investigation 3-F on page 258

Activity

Energy Transformation in Other Earth Processes

1. Choose one of the processes below or another of Earth's processes that interests you.
 - life processes (photosynthesis and cellular respiration)
 - geological processes (volcanic activity and the motion of Earth's crusts)
 - tides
 - winds
2. Determine the role energy transformation plays in the process. Use the resources available to you (this textbook, group discussion, online or print resources, and/or resources provided by your teacher) to do so.
3. Share your explanation with the class by creating an info-graphic using either self-produced artwork or technology.



Before you leave this page . . .

1. Describe the role energy transformation plays in the water cycle.
2. What is specific heat capacity?
3. Why are coastal temperatures more moderate than inland ones?

CONCEPT 4

Energy transformation and transfer can harm aquatic and terrestrial ecosystems.

Energy transformations that can harm the environment may be part of natural events, such as forest fires or volcanic activity. However, great harm is also, often, the result of human activity. These two pages explore two examples.

Aquatic Ecosystems and Climate Change

Aquatic ecosystems are especially harmed by changes in temperature. Earth's average temperature is about 1°C warmer than it was last century due to the human-induced greenhouse effect. Greenhouse gases play a large role in warming Earth's atmosphere. For at least 200 years, humans have been adding significant amounts of these gases to the atmosphere. This has been mainly to generate electrical energy, power vehicles, warm homes, and carry out industrial processes.



Many aquatic organisms thrive at a certain water temperature. As ocean temperature increases, many species move to cooler waters. This affects the animals that rely on them. In B.C., warming ocean water is affecting the tufted puffin. The small fish the puffins eat are moving to cooler waters. Mature puffins follow their food source and abandon their young in the process. Each year, fewer young puffins survive to adulthood.



Warming ocean temperatures are also destroying an important nursery and habitat for ocean life: coral reefs. Corals are animals that live in close association with algae. Through photosynthesis, the algae provide the coral with food. This relationship is threatened by climate change. As oceans warm, the corals expel the algae and die. In the process, they become white, or bleached.

Connect to Investigation 3-G on page 261

Extending the Connections

Terrestrial Ecosystems and Climate Change

Develop and investigate questions about the effects of climate change on terrestrial ecosystems. How can you use your findings to educate people about the issues involved?

Terrestrial Ecosystems and Radiation Exposure

Terrestrial ecosystems are much more likely to be harmed by radiation exposure than aquatic ones. This is because radioactive materials accumulate in soil, while they tend to flush through aquatic ecosystems more quickly. Radioactive materials can build up in terrestrial ecosystems due to accidents at nuclear power plants. Although such accidents are infrequent, their effects can be devastating to ecosystems. Improperly stored nuclear wastes can also contaminate terrestrial ecosystems. This issue is explored on the next page.



In March 2011, radioactive materials were released into the environment after the Fukushima Daiichi nuclear plant in Japan was struck by a tsunami. Harmful effects of the radiation were seen within two months of the accident. Scientists found mutations in eye and wing structures of butterflies near the accident site. The butterflies had been overwintering as larvae when the accident occurred. Butterflies studied six months after the accident had even more mutations. This showed that mutations in the butterflies' reproductive cells were being passed to their offspring.



In 1986, a nuclear reactor at the Chernobyl nuclear plant ruptured when it overheated. The accident released part of the radioactive reactor core into the surrounding area. Radioactive materials were still being released 10 days later. Plants and animals within 20–30 km of the reactor were the most affected by radiation from the nuclear decay of the materials. In the nearby Red Forest, all the trees were killed. Over 30 years later, the effects of radiation poisoning are still felt in the area. Trees do not grow as quickly. Birds have smaller brain sizes. Perhaps most important is the loss of fungi, microorganisms, and insects involved in biological decay. This decay frees nutrients in dead matter, so they can be taken up by living things. In the Red Forest, many trees did not decay even 15 to 20 years after they were killed.



Before you leave this page . . .

1. Explain how climate change is affecting at least one aquatic organism.
2. Why are radioactive materials so harmful to terrestrial ecosystems?

Storing Nuclear Waste Safely

What's the Issue?

Nuclear wastes contain radioactive isotopes that release harmful radiation through nuclear decay over time. Eventually they become less radioactive. However, some nuclear waste takes a very long time to reach this state. Because there are no nuclear power plants in British Columbia, the issue of safe storage of nuclear wastes may seem very far from home. However, improper storage of these wastes can harm ecosystems around the world.

Nuclear energy provides about 15% of Canada's electrical energy. The safe management and storage of nuclear wastes is overseen by government organizations and regulatory bodies. Nuclear wastes are classified based on how much containment and isolation they require to be stored safely.

- **Low-level and Intermediate-level Waste** These are stored at the sites where they are produced. Once their radioactivity decays to acceptable levels, they are disposed of by conventional means, such as in landfills.
- **High-level Waste** These include used nuclear fuel and other waste from producing nuclear power. It can take a long time to undergo nuclear decay. The waste also gives off a lot of thermal energy, so it must be handled in two phases. First, used fuel bundles are stored under water in secure, leak-proof pools for 6 to 10 years until they cool. Then the bundles are transferred to concrete containers where they decay further.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. Compare nuclear waste with another harmful form of pollution. Which do you think is a greater threat to ecosystem health? Explain.
2. Use resources available to you, including those provided by your teacher, to find out more about nuclear waste contamination at the Hanford Nuclear Reservation in Washington. What mistakes were made at the site in terms of safe waste storage? Describe any harmful effects to the environment that resulted from these mistakes.

What's the Issue?

In 2017, 20 marine scientists wrote a letter to the federal government on the behalf of 78 B.C. residents who couldn't write it themselves—the endangered orcas that make their home in southern B.C.'s Salish Sea. The Salish Sea includes the Strait of Georgia, the Strait of Juan de Fuca, and Puget Sound. These waters are busy marine shipping routes. Many tankers and freighters pass through them on their way to the ports of Vancouver, B.C. and Seattle, Washington.

In their letter, the scientists called on the government to commit to reducing underwater noise pollution, such as that produced by marine vessels, in the region. As explained in the letter, noise pollution interferes with the orcas' ability to communicate and find food. It is also a cause of chronic stress. The urgency of the letter is underscored by the fact that the Salish Sea will get louder if proposed oil and coal terminal expansions go ahead in the area.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. Freighters produce much of the underwater noise in the Salish Sea. These ships don't have to make as much noise as they do. Engine and propeller noise can both be reduced. In fact, some naval vessels are designed to be extremely quiet. Despite this, little effort has been made to reduce engine and propeller noise. Why do you think this is the case?
2. Do you think noise pollution should be regulated to protect endangered animals in marine environments? Prepare an argument or participate in a debate in which you argue one side of the issue.
3. What other animals are harmed by noise pollution? How are they affected by the noise? Use the information provided by your teacher, or research an animal of personal interest.

Make a Difference

Marilyn Slett: Advocating for an Indigenous Marine Response Centre

Marilyn Slett is Chief Councillor of the Heiltsuk Nation. She is a vocal advocate for the need for an improved oil-spill response plan in B.C. marine ecosystems. This need was reinforced in October 2016, when a tug boat crashed near Bella Bella in an endangered abalone habitat, running aground onto the shores of an ancient clam garden for the Heiltsuk. The boat released diesel fuel, a harmful marine contaminant, into the water. According to an article the Chief Councillor wrote in 2017, the response to this spill was unsafe and poorly managed. Clearly, there was need for improvement. This improvement has come in the form of the Heiltsuk proposal for an Indigenous Marine Response Centre.

First Peoples are often the first responders to marine emergencies in B.C. Creating a response centre near Bella Bella would allow First Peoples to respond to a spill on the central and north coasts of the province within 5 hours

or less. The centre would use both Western and First Peoples knowledge to mount the effective response. According to Marilyn Slett, “The question is how best to develop community capacity to address emergencies in our territories as they arise, and how to deliver the world class protection Canada’s oceans and communities need and deserve.”

This Indigenous Marine Response Centre may provide a much-needed solution.



Evaluate, Apply, and Communicate

1. What lessons can you learn from Marilyn Slett about preparing for and dealing with emergencies that threaten the health of your community?
2. Mobilize your community to assess emergency response plans or to develop plans if they are lacking.



Check Your Understanding of Topic 3.3

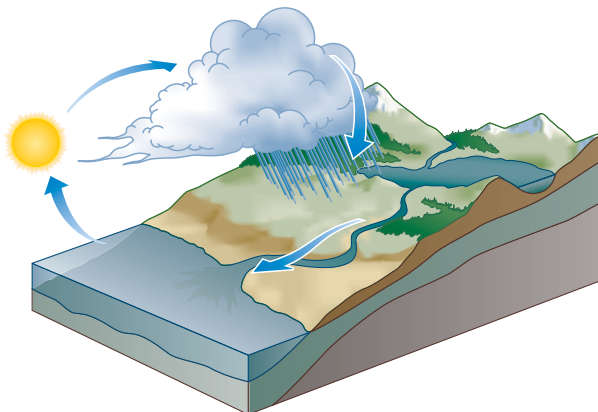
QP Questioning and Predicting
 PC Planning and Conducting
 PA Processing and Analyzing
 E Evaluating
AI Applying and Innovating
C Communicating

Understanding Key Ideas

1. Consider the photo below. PA AI



- If Earth, including its atmosphere, is defined as the system, what are the surroundings?
 - How does this system absorb energy from its surroundings?
 - How does this system release energy to its surroundings?
2. Identify the three main types of energy that play a role in Earth's processes. PA AI
3. Refer to the stages of the water cycle below to describe how energy moves between Earth's surface and atmosphere. PA C



- What is the greenhouse effect?
 - Why is it important to life on Earth? PA AI
- Use a graphic organizer to explain the roles that absorption, radiation, conduction, and convection play in warming the atmosphere. PA C
- Why are terrestrial ecosystems harmed more by nuclear radiation than aquatic ecosystems are? PA AI

Connecting Ideas

- Imagine that you move from Vancouver to Kamloops. Predict how and explain why the average temperatures you experience will now be different in winter and in summer. QP PA AI
- Design the procedure for a controlled experiment that could be used to determine the effects of warming oceans on coral reefs. PC PA AI

Making New Connections

- Climate change also affects ocean temperatures. As the temperature of water increases, the space between the water molecules expands. This causes sea level to rise. How do you predict that aquatic and terrestrial ecosystems would be affected by these changes? Explain your reasoning. QP PA AI
- One molecule of nitrous oxide warms Earth's atmosphere more than one molecule of carbon dioxide. And yet, new reports about global warming and climate change focus substantially on carbon dioxide and rarely mention nitrous oxide or other greenhouse gases that are more potent than carbon dioxide. Why might this be the case? Provide reasons to justify your opinion. E AI C

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing

Safety



- Use caution when handling the lamps, as the bulbs will become very hot.

What You Need

- 2 heat lamps
- 2 500 mL beakers
- 4 thermometers
- 125 mL water
- 125 mL soil
- apparatus to support heat lamps
- graph paper
- 2 data table templates

Specific Heat Capacity

PART A: Comparing Temperature Changes

In this part, you will compare the ability of water and soil to absorb radiant energy, and its affect on overlying air temperature.

Question

How can you compare the absorption of radiant energy by water and soil, and its affect on overlying air temperature?

Procedure

1. Your teacher will set up the apparatus in the photo and provide the mass of the water and soil.
2. Obtain copies of the two data tables from your teacher.
3. Turn on the heat lamps. Record the temperature at each of the four thermometers at one-minute intervals for a total of 10 min.
4. Turn off the heat lamps, and wait a few minutes to let the heat dissipate.
5. With the lamps off, repeat the temperature readings from step 3.
6. Make two graphs using the results from your data tables. Use a different symbol or colour for each variable.



Analyze

1. What did you notice about the temperature changes in the soil compared to those measured in the water
 - a) when the lamp was on?
 - b) when the lamp was off?
2. What did you notice about the temperature changes above the soil compared to those measured above the water
 - a) when the lamp was on?
 - b) when the lamp was off?

Conclude and Apply

3. Which has a higher specific heat capacity: soil or water? Explain your choice.
4. Sea breezes are local winds common to coastal areas during daytime hours. They are also called onshore breezes. Land breezes are local winds that occur during nighttime hours. They are also called offshore breezes. Use what you have learned in this Investigation to explain how sea breezes and land breezes form. Include diagrams in your answer.

PART B: Calculating Energy Transformation

In this part, you will learn how to carry out calculations involving the relationships among energy, mass, and changes in temperature.

Question

How can you solve problems involving relationships among energy, mass, and changes in temperature?

Procedure

1. Read the text below about calculating specific heat capacity. Then study the sample problems.

Scientists have used many measurements to determine the values for the specific heat capacity, c , of many materials. A few of these values are listed in the table.

Table of Specific Heat Capacities for Some Common Substances

Substance	Specific Heat Capacity (J/g°C)
Water (liquid)	4.18
Dry soil	0.800
Wet soil	1.48
Wood (pine)	1.50
Steel	0.490
Salt (sodium chloride)	0.880
Air (dry)	1.005

Scientists have also determined that the mathematical relationship between heat absorbed by a material and its specific heat capacity, mass, and temperature change can be summarized in the following equation.

$Q = mc\Delta T$, where Q represents the amount of heat absorbed or released by a substance of mass, m , with a specific heat capacity, c , at a change in temperature of ΔT .

Sample Problems

- A.** Using data from the table, determine the heat that must be added to raise the temperature of a cup (250 g) of water from 20°C to 100°C.
- B.** City engineers want to know the specific heat capacity of a new sample of asphalt. A chemist set up a system that would add 3.0×10^4 J of thermal energy to 1250 g of the sample. The chemist measured the change in the temperature of the asphalt as 26°C. What was the specific heat capacity?

Solutions

- A.** The change in temperature is
 $\Delta T = 100^\circ\text{C} - 20^\circ\text{C} = 80^\circ\text{C}$.

Use the equation to find the heat needed.

$$\begin{aligned} Q &= mc\Delta T \\ &= 250\text{ g} \times 4.18 \frac{\text{J}}{\text{g}^\circ\text{C}} \times 80^\circ\text{C} \\ &= 8.36 \times 10^4 \text{ J} \\ &\cong 8 \times 10^4 \text{ J} \end{aligned}$$

- B.** Rearrange the equation to solve for c . Then substitute the values into the equation.

$$\begin{aligned} Q &= mc\Delta T \\ \frac{Q}{m\Delta T} &= \frac{mc\Delta T}{m\Delta T} \\ c &= \frac{Q}{m\Delta T} \\ c &= \frac{3.0 \times 10^4 \text{ J}}{1250 \text{ g} \times 26^\circ\text{C}} \\ c &= 0.923 \frac{\text{J}}{\text{g}^\circ\text{C}} \\ c &\cong 0.92 \frac{\text{J}}{\text{g}^\circ\text{C}} \end{aligned}$$

2. Complete the following practice problems.
- Compare the amount of heat necessary to raise the temperature of one cup (250 g) of water and one bathtub (3.03×10^5 g) full of water, from 20°C to 50°C.
 - How much would the temperature of 250 g of salt increase if 8.36 J of energy were added to it? How does this compare to the temperature increase of water under the same conditions?

Process and Analyze

- Use data from Part A to calculate the amount of heat absorbed by the water.
- If the soil absorbed approximately the same amount of heat as the water, calculate the specific heat capacity of the soil.
- Do you think that the soil actually absorbed approximately the same amount of heat as the water absorbed? Explain your reasoning.
- Consult the table for the specific heat capacity of dry soil. Is your calculated value similar to the appropriate value given here? If not, does the specific heat capacity of dry soil in the table indicate that the soil actually absorbed more or less heat than water in the Investigation? Explain.

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

Safety



- Use caution when handling heat sources such as a lamp, since the light bulb will become very hot.
- Advise your teacher if you are allergic to eggs.

What You Need

Part A:

- materials as per your group's design

Part B:

- shells from 3 hard-boiled eggs
- 3 containers with lids
- weak vinegar solution
- strong vinegar solution
- water
- universal pH paper
- scale

Modelling Changes in Earth's Systems

Scientists often create physical models of systems they are studying, including Earth's systems. In this Investigation, you will create models that will help you understand how human-induced changes can affect these systems.

PART A: Modelling Changes to Earth's Atmosphere

In Part A, you will model and test the natural greenhouse effect. Then you will adjust your design to model and test the human-induced greenhouse effect.

Question

How can you model and test the natural and human-induced greenhouse effect?



Problem

You must design and build a physical model of the natural greenhouse effect. Your model should meet the following criteria:

- allow light or solar energy to enter the system
- absorb the solar or light energy
- transform the solar or light energy into thermal energy
- radiate the thermal energy as infrared radiation
- prevent infrared radiation from leaving the system
- allow you to measure the temperature of your system
- allow you to adjust variables to model the human-induced greenhouse effect

Procedure

1. With your group, review the problem. Identify the variables in your investigation and how you can control them. Decide how you will compare the temperature inside your model with the temperature outside your model.
2. Brainstorm materials that you could use and how you could use them. Create a materials list. Decide on any safety precautions that are required.
3. Sketch a diagram of your model.
4. Describe your model to another group. Incorporate any appropriate changes they suggest into your design and sketch.
5. Write a plan describing your model and how you will build it. Include your materials list and safety precautions. Have your teacher approve your plan.
6. Build your model. Troubleshoot any problems as they arise.

7. Plan a procedure to test your greenhouse model and compare the results to those measured outside your model. Remember to account for and control all the variables, except the one you are measuring.
8. Create a table to record your data.
9. Carry out your test. Record your data.
10. Adjust variables to model the human-induced greenhouse effect.
11. Repeat steps 8 and 9 for your adjusted model.

Process, Analyze, and Communicate

1. Graph your temperature data from inside your model of the natural greenhouse effect, outside your model, and after adjusting variables to model the human-induced greenhouse effect.
2. Compare your graphs. Summarize the results of your comparison. Did you get the results you expected? Explain.
3. Share your model and your summary with the class.

Evaluate

4. a) In what ways was your model an accurate representation of the natural greenhouse effect? the human-induced greenhouse effect?
b) In what ways was your model an inaccurate representation of these two effects?
5. Based on the models of other classmates, how could you refine your design to model the greenhouse effect more accurately?

PART B: Model Ocean Changes

Oceans absorb carbon dioxide from the atmosphere. This helps moderate atmospheric temperature increase due to the human-induced greenhouse effect. However, the more carbon dioxide oceans absorb, the more acidic they become. This is because carbon dioxide reacts with water to form carbonic acid. If ocean water gets too acidic, animals such as corals, clams, and sea urchins cannot build their shells or skeletons very well. In this investigation, you will model and test how shells that contain calcium are affected by different levels of acidity.



Question

How can you model and test how shells containing calcium are affected by different levels of ocean acidity?

Procedure

1. In this investigation, egg shells are used to model sea shells. Your group will test what happens when the shells are placed in water with different levels of acidity. Read through all the steps. Make and record predictions about what you will observe.
2. Measure and record the pH of the three liquids.
3. Divide the shells into three piles. Measure and record the mass of each pile.
4. Do the following for each liquid.
 - a) Place the shells in a container.
 - b) Pour enough liquid to just cover the shells.
 - c) Secure the lid, and place the container where it will be left alone for one day.

5. The next day, carefully remove the shells from each container. Record any changes you observe in the shells.
6. Measure and record the mass of the shells from each container.
7. Measure and record the pH of each liquid.

Process and Analyze

1. Describe the changes you observed in the shells.
2. Which shells changed the most? How did this compare with your predictions?

Apply

3. Use your results to explain how an animal with a calcium shell might be affected when exposed to ocean water that is too acidic.